

CS361 – Computer Architecture

Week 2 – Q&A; Activity Sheet

Section 1: Conceptual Recall

- 1. What is the difference between a core and a thread?
- 2. How many threads can a quad-core CPU with hyper-threading run?
- 3. Define latency and bandwidth in your own words.
- 4. Why must an Instruction Set Architecture (ISA) survive rapid technology changes?

Section 2: Application

- 1. If a CPU has 8 physical cores and supports hyper-threading, how many logical cores does it show to the OS?
- 2. If bandwidth improves, which benefits more multithreaded programs or single-threaded programs? Why?
- 3. Suppose a memory access takes 100 ns (latency). If bandwidth doubles but latency remains the same, what improves and what stays constant?
- 4. If Moore's Law slows down, what other techniques can designers use to keep improving performance?

Section 3: Critical Thinking

- 1. Why doesn't adding more cores always improve latency?
- 2. Which applications benefit more from bandwidth growth: gaming, video streaming, or database queries? Why?
- 3. If latency improves 2×, bandwidth might improve 4×. Why is it easier to scale bandwidth than latency?
- 4. Imagine you are a designer: would you prioritize performance-per-watt or raw performance for mobile devices? Why?